

WELCOME TO

SIT5



Deeper Dive Into Software

Tuesday 9/9



Debrief

Main goal of this meeting:

- Dive deeper into computer software & coding
- Discuss our next meeting
- Answer questions and receive feedback



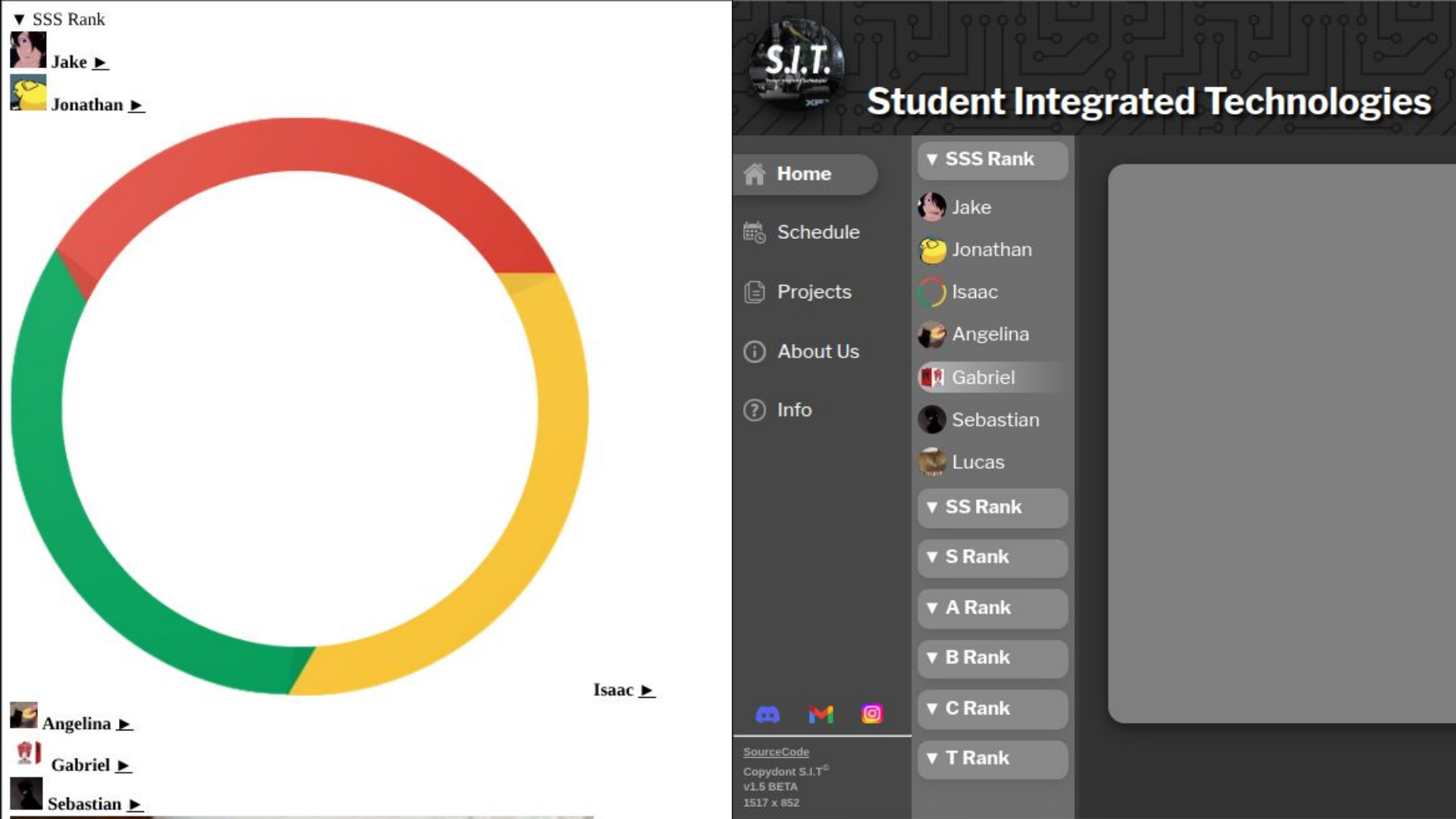
Coding Principles

- We will be introducing the following languages today:
 - HTML
 - CSS
 - JAVASCRIPT
 - PYTHON
- We will simply be sharing their concepts and syntax for a better understanding of their functions
- Consider AP Computer Science Principles or AP Computer Science A (2nd year) as a future class if you are interested in learning about coding



coding Principles: HTML & CSS

- Hypertext Markup Language is a markup language, it defines “tags” for different “elements” that are placed on screen – such as paragraphs, images, etc.
- It’s generally not considered a coding language, although it’s the building block for the web
- Cascading Style Sheet is a considered a style sheet rather than a coding language. It defines the colours and shapes that you see on a website
- CSS also sets margins, moves elements, and defines all of the non-information you see on a website





Coding Principles: JS & PY

- Javascript and Python are both considered coding languages and are the most popular, but they both differ
- Javascript is the only language that is natively supported by browsers, and is most commonly used in websites
- Javascript does not need indentation to properly run
- Python is not natively supported by default, and is most commonly used for applications and games

• Python needs indentation to run / define blocks of

Line wrap

```

1 <html>
2 <head>
3   <link rel="shortcut icon" href="icons-images/favicon.png" type="image/x-icon">
4   <title>8pairs - Landing</title>
5   <link rel="stylesheet" href="style.css" />
6   <meta name="description" content="The official project site for 8pairs. Most of the code is written in JavaScript." />
7   <meta property="og:image" content="https://8pairs.net/icons-images/8pairs.png" />
8   <noscript>You don't have JavaScript enabled.</noscript>
9 </head>
10 <body>
11 <!-- [REDACTED] -->
12 <div class="main">
13   <div class="left">
14     <div class="profile">
15       
16       <p>8p8c</p>
17       <p class="smallify">"Suffering from constant electrocution"</p>
18       <p class="smallify">
19         <a href="https://github.com/8pairs" target="_blank">GitHub</a>
20         <a href="https://discord.com/users/1136008749920817192" target="_blank">Discord</a>
21       </p>
22     </div>
23   </div>
24   <div class="header">
25     <div class="header-main">
26       <p>Student Integrated Technologies</p>
27       <hr>
28     </div>
29     <div class="inner">
30       <div class="project-container" onclick="if (!window.__cfRL__){}>
31         <p class="title">SIT&nbsp;&nbsp;&nbsp;<span class="description">
32       </div>
33     </div>
34   </div>
35   <p class="opacity">Projects</p>
36   <hr class="opacity">
37   <div class="inner opacity">
38     <div class="project-container" onclick="if (!window.__cfRL__){}>
39       <p class="title">your&nbsp;&nbsp;&nbsp;<span class="description">
40     </div>
41   </div>
42   <div class="project-container" onclick="if (!window.__cfRL__){}>
43     <p class="title">enc&nbsp;&nbsp;&nbsp;<span class="description">
44   </div>
45 </div>
46

```

```

@import url('https://fonts.googleapis.com/css2?family=Rubik:ital,wght@0,300..900;1,300..900');

body {
  background-color: #000;
  font-family: "Rubik", sans-serif;
  color: #d3869b;
}

::-webkit-scrollbar {
  width: 12px;
}

::-webkit-scrollbar-thumb {
  background: #d3869b;
  border-radius: 6px;
}

.opacity {
  opacity: 0.5;
}

.main {
  margin: auto;
  border-radius: 20px;
  border: 4px #d3869b solid;
  height: 80%;
  width: 80%;
  position: absolute;
  top: 50%;
  left: 50%;
  transform: translate(-50%, -50%);
  box-shadow: 0 0 50px 3px #d3869b;
}

>.left {
  display: flex;
  align-items: center;
  justify-content: center;
  float: left;
  width: 35%;
}

>.profile {
  text-align: center;
  width: fit-content;
  margin-top: 60px;
}

>img {

```

```

1 window.onload = () => {
2     homeBTN.setAttribute('id', 'selected')
3 }
4
5 const ifrm = document.getElementById('main-iframe');
6
7 function siteHome() {
8     ifrm.setAttribute('src', 'home.html');
9 }
10
11 function siteSchedule() {
12     ifrm.setAttribute('src', 'schedule.html');
13 }
14
15 function siteProjects() {
16     ifrm.setAttribute('src', 'projects.html');
17 }
18
19 function siteAboutUs() {
20     ifrm.setAttribute('src', 'about-us.html');
21 }
22
23 function siteAp() {
24     ifrm.setAttribute('src', 'agents.html');
25 }
26
27 function siteInfo() {
28     ifrm.setAttribute('src', 'info.html');
29 }
30
31 const homeBTN = document.getElementById('home')
32 const scheBTN = document.getElementById('schedule')
33 const projBTN = document.getElementById('projects')
34 const abtBTN = document.getElementById('about-us')
35 const ap = document.getElementById('agent-p')
36 const info = document.getElementById('info')
37
38 function clkHome() {
39     homeBTN.setAttribute('id', 'selected');
40
41     scheBTN.setAttribute('id', 'schedule');
42     projBTN.setAttribute('id', 'projects');
43     abtBTN.setAttribute('id', 'about-us');
44     ap.setAttribute('id', 'agent-p')
45     info.setAttribute('id', 'info')
46 }

```

```

1 import os, time
2 x = "Welcome to The Mirror."
3 y = 0
4
5 while y <= len(x):
6     os.system("clear")
7     print(x[:y])
8     time.sleep(0.2)
9     y = y+1
10 time.sleep(2)
11 x = "You will stay here for as long as you can, you may
12 forfeit when you would like to."
13 y = 0
14
15 while y <= len(x):
16     os.system("clear")
17     print(x[:y])
18     time.sleep(0.2)
19     y = y+1
20 time.sleep(2)
21 x = "Please stand here, and stare into the mirror."
22 y = 0
23
24 while y <= len(x):
25     os.system("clear")
26     print(x[:y])
27     time.sleep(0.2)
28     y = y+1
29 time.sleep(2)
30 x = "This is you in the mirror:"
31 y = 0
32
33 while y <= len(x):
34     os.system("clear")
35     print(x[:y])

```


Navigating Files

File Explorer

- Easy to navigate and view
- All tools are a click away
- Very easy to copy/move files

Terminal / CMD

- More complicated and possibly confusing
- Tools need to be executed with commands such as:
 - move
 - copy
 - mkdir
 - cd / dir
 - del



Program Installation

- All programs are executed via .exe or .msi files (even if they are invoked by an install button inside of another program)
- Programs are all automatically installed in directories like C:/Program Files or a user specified path
- Some programs require administrator privileges to be able to install into different directories/paths or execute higher level actions such as restarting your computer or editing system data
- Fun fact: Most default Windows programs are installed in either C:/Windows, or C:/Windows/System32.



All OSs share:

- **Background programs; Programs that start and run in the 'background' automatically (generally very necessary)**
- **A "Kernel" that manages memory and CPU use. The very basic component of an operating system that runs below all programs and is generally not editable**
- **A terminal where commands can be executed. Commands vary but you may see similar ones between operating systems**
- **Manage resources such as files, input/outputs, HID devices, visual I/O as well as executing programs**



Extra Information

- All coding languages are different, but most of them follow the same general syntax such as:
 - Indentations
 - variables
- There are two types of modern applications within windows. 32-bit and 64-bit
- Fun fact: You can put “ view-source: ” in front of a URL to view its raw HTML code



Next Meeting

Tuesday 9/16

Ground plan:

- To Be Determined
(Break Upcoming)

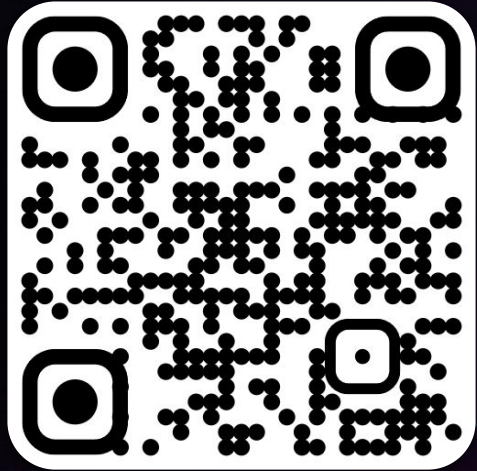


We want to know what YOU want to learn
about
next Tuesday!



Info

Discord



Website



Instagram

